

Zhi FANG

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RESEARCH INTERESTS

Deep Learning; Graph Representation Learning, Brain Imaging and Intelligent Diagnosis; Large Language Model;

EDUCATION

University of Electronic Science and Technology of China (Project 985)

Chengdu, China

Master of Engineering in Biomedical Engineering

Sep 2024 - Jun 2027 (Expected)

- **GPA:** 3.43/4.0 | **Average Score:** 83.09/100

Chengdu University of Information Technology

Chengdu, China

Bachelor of Engineering in Communications Engineering

Sep 2020 - Jun 2024

- **GPA:** 3.51/4.0 | **Average Score:** 87.69/100

PUBLICATIONS

- Liu, H.†, Zhang, J.†, **Fang, Z.†**, Jiang, X.*, Huang, W.*, Li, R.*. **Multi-hop spatio-temporal graph convolutional networks for brain disorder diagnosis and prognosis.** (Accepted by *Pattern Recognition*)
- **Fang, Z.**, Chen, H.*, Huang, W.*, Li, R.*. **Multi-Scale Feature Fusion Networks for Identifying Abnormal Brain Networks.** (Under review by *IEEE TNNLS*)
- Han, H.†, Liu, Q.†, Sun, X., **Fang, Z.**, Li, R.*. **Ablation Surgery Modulates a Common Functional Network in Focal Epilepsy: Disconnection, Recovery and Reorganization.** (First revision by *Neurology*)
- **Fang, Z.**, Chen, H.*, Li, R.*. **K-MIND: A Knowledge-guided Multi-modal Interaction Network for Brain Disorders** (Manuscript in Preparation)

RESEARCH EXPERIENCE

Multi-hop spatio-temporal graph convolutional networks for brain disorder diagnosis and prognosis.

Core Member | Supervisor: Rong Li, Huaifu Chen

Funded by Chinese Brain Project (Grant No. 2022ZD0208903).

Dec 2024 – Apr 2026

- **Problem:** Existing GNN attention methods only aggregate one-hop neighbors, ignoring long-range indirect connections in sparse brain networks, while multi-site fMRI heterogeneity limits cross-site generalizability.
- **Method:** Proposed MSTGCN, extending spatial attention to multi-hop connections via geometric-distribution-weighted attention diffusion; employed a Transformer encoder for dynamic FC temporal modeling and RevCon regularization to suppress site-specific biases by maximizing intra-class diversity.
- **Outcome:** Achieved ACC/AUROC 85.52%/0.913 (Site 1), 82.09%/0.869 (Cross-site), 69.23%/0.741 (ABIDE), outperforming 13 SOTA; surgical prognosis 82.0% ACC.

Multi-Scale Feature Fusion Networks for Identifying Abnormal Brain Networks

Team Leader | Supervisor: Rong Li, Huaifu Chen

Presented at 2025 ISICDM & 2026 Chinese Intelligent Health & Bioinformatics Conference

Funded by National Natural Science Foundation of China (Grant No. 82372085, 62333003) **Dec 2025 – Present**

- **Problem:** Existing brain network models analyze static and dynamic FC at a single scale, failing to capture hierarchical spatiotemporal dependencies, while multi-site heterogeneity limits cross-site generalizability.
- **Method:** Proposed MSD-Net, integrating group-guided contrastive pooling (static) and diffusion spatial attention (dynamic) via cross-view fusion, with anatomical-functional dual-scale tokenization and local-global fusion, optimized by RevCon and entropy-based sparsity regularization.
- **Outcome:** Achieved 85.00%/74.29%/81.16% ACC on three TLE private datasets and 73.77%/71.03% on ABIDE-I/ADHD-200, outperforming 13 SOTA; saliency maps aligned with neuroscience findings.

A Multimodal Knowledge-Guided Fusion Framework for Brain Disorder Diagnosis

Team Leader | Supervisor: Rong Li, Huaifu Chen

Funded by National Natural Science Foundation of China (Grant No. 82372085, 62333003) **May 2026 - Present**

- **Problem:** Existing multimodal fusion methods coarsely concatenate fMRI and sMRI features without incorporating medical knowledge to guide cross-modality interaction.
- **Method:** Proposed a knowledge-guided multimodal framework encoding PubMedBERT-derived semantic priors for 246 brain subregions, fusing fMRI and sMRI connectivity via structure-guided projection, orthonormal clustered attention, and QFormer cross-modal interaction.

- **Outcome:** Evaluated on 2,350 subjects from 3 private TLE datasets and two public datasets (ABIDE, ADHD-200). Achieved ACC of TLE 84.8% (Best performance of 3 centers), ADHD 76.5%, ASD 70.7%.

Mobile Agent — LLM-based Multi-step Autonomous Decision-Making System

Independent Researcher

Oct 2025 - Feb 2026

- **Problem:** Conventional voice assistants are limited to single-turn commands and cannot handle cross-scenario, long-horizon tasks requiring multi-step reasoning and tool use.
- **Method:** Designed an MLP-based Progress Reward Model for step-wise reward back-propagation to address sparse-reward convergence in long-horizon tasks; introduced environment Grounding Reward for real-time hallucination interception and operation-compliance reinforcement; built a Step-level PPO RL pipeline combining task-progress gain and operation-compliance dual rewards.
- **Outcome:** Validated on WebShop, ALFWorld and VirtualHome datasets; task success rate improved from 82% to 90%; interaction time reduced 28% (25s → 18s); human takeover < 8%.

Agentic RAG — Intelligent Epilepsy Consultation & Retrieval System

Independent Researcher

Mar 2026 - May 2026

- **Problem:** Epilepsy diagnosis relies on clinician experience and suffers from inefficient literature retrieval, weak semantic understanding, and poor integration of published evidence with clinical cases.
- **Method:** Designed parent-child chunking with parent id backtracking and LLM three-way intent routing to suppress noisy recalls; implemented BGE-M3 hybrid retrieval with Multi-Query rewriting, improving recall 15–20%; built StateGraph multi-node workflow with dynamic intent-driven branch switching.
- **Outcome:** Integrated 300+ epilepsy publications and 2,000+ clinical assessment items; RAGAS + LLM-as-a-Judge dual evaluation achieved 85%+ accuracy; average response < 5s (p95 < 7s).

PATENTS & SOFTWARE COPYRIGHT

- Platform for Evaluating the Efficacy of Epilepsy Surgery Based on a Digital Brain Model, Version 1.0, 2025SR0523331, Chinese Software Copyright (filed), 2025. (Listed as 3rd of 3 inventors)
- A Method for Identifying Epileptic Abnormal Brain Networks Based on a Multiscale Static-Dynamic Fusion Network, CN 2025115041286, China Patent (filed), 2025. (Listed as 2nd of 3 inventors)
- Epilepsy Brain Imaging Consortium Multicenter Data Sharing Platform [Abbreviation: EBIC] V1.0, 2026SR0397180, Chinese Software Copyright (filed), 2026. (Listed as 3rd of 4 inventors)

WORK & INTERNSHIP EXPERIENCE

Lenovo, Solutions & Services Group, IT Delivery Team

AI Application Development Engineer, Intern

Jun 2026 - Sep 2026

- Developed end-to-end AI solutions for CSD-AI, Lenovo's global channel sales data quality platform serving NA/LA/AP regions; built LLM-based intent recognition models for automated text and email anomaly detection; delivered the full development-to-production lifecycle from prototyping through deployment and release.

The Clinical Hospital of Chengdu Brain Science Institute

MRI Machine Operator, Intern

Apr 2025 – Apr 2026

- Coordinated acquisition of 3.0T MRI and 64-channel EEG, delivering a task-driven “EEG-fMRI synchronous acquisition” multimodal dataset; preprocessed motion correction, denoising, and initial FC; delivered analysis-ready data.

HONORS & AWARDS

Scholarships

- First-Class Scholarship (Top 10%), University of Electronic Science and Technology of China **2024, 2025**
- First-Class Scholarship (Top 5%), Chengdu University of Information Technology **2023, 2022, 2021**
- The Outstanding Postgraduate Student honor by UESTC for the year 2025
- The Outstanding Undergraduate Student of Sichuan province and CUIT for the year 2024

Academic Competitions

- First Prize at Research English Speech Contest for Chinese College Students; **2025**
- Second Prize at the Provincial Level in the International College Students' Innovation Competition **2025**
- First Prizes in Four Provinces at the National College Students Mathematical Modeling Competition **2022**
- Third Prize at China University Computer Competition—AI Innovation Contest National Finals **2022**

LANGUAGES & SKILLS

- **Languages:** Chinese (Native); English (IELTS 6.5)
- **Technical Skills:** Python, PyTorch, vLLM, Hugging Face, R, EndNote, Overleaf, MATLAB, Git, SQL